

★ WEEK 11 · PART 3 · PRACTICE

OU in Trading, Worked by Hand

Screen Pairs by Half-Life; Turn Reversion Into Return

Stage V - the economics of a mean-reversion portfolio

Part 3's Stage V, with the arithmetic shown. We use the half-life as a filter to screen candidate pairs, then compute the trade economics - how a faster reverter recycles capital into more annual return from the same per-trade edge.

3P

PART 3 · APPLICATIONS

From One Pair to a Portfolio

Case A: screen four candidate pairs by half-life, keeping only fast, stationary reverters. Case B: the trade economics - how the half-life drives turnover and annual return. Then the five-stage journey and a Claude Code lab.

What Part 3 Told Us

Now we screen and size it - with numbers

1

OU replaces heuristics with a model

Instead of ad-hoc thresholds, the spread's own statistics set the bands - principle where there was guesswork.

2

The half-life is a trade filter

It tells you how fast a pair reverts, so you can screen candidates and keep only the ones worth trading.

3

Re-estimate and stay humble

Spreads drift and break. A fitted half-life is only as good as the relationship that produced it.

Everything here is a calculator exercise: four candidate pairs, one filter, two economics.

How These Cases Work

Data, then arithmetic, then meaning

Each case has three steps

- Start from fitted AR(1) coefficients.
- Screen by half-life or size the trade.
- Then read the portfolio decision.

What you should watch

- How the filter rejects slow and non-reverting pairs.
- How the half-life drives trades per year.
- How turnover turns the same edge into more return.

Screen Pairs by Half-Life

Not every spread is worth trading

You have four candidate pairs, each with a fitted AR(1) coefficient. The half-life turns each into a single, comparable number - how fast it reverts. A good trade needs a half-life short enough to turn capital, but the pair must actually revert. We screen all four by hand

The filter

- Accept a half-life between 2 and 30 days.
- Too slow ties up capital for little return.
- The coefficient must be below one to revert.

What we'll do by hand

- Convert each coefficient to a half-life.
- Apply the filter to all four.
- Keep the tradeable ones.

The Half-Life Filter

One number to screen a pair

$$\text{half-life} = \frac{\ln 2}{-\ln \phi}, \quad \text{ACCEPT if } 2 \leq \text{HL} \leq 30 \text{ days and } \phi < 1 \text{ (stationary)}$$

The upper bound

- A very long half-life is a capital trap.
- The spread reverts too slowly to be worth it.
- Reject anything above about 30 days.

The hard floor

- A coefficient at or above one doesn't revert.
- There's no half-life - it's non-stationary.
- Reject it outright, whatever else it shows.

Four Pairs, Four Half-Lives

Screen the candidates

pair	AR(1) phi	kappa = $-\ln(\text{phi})$	half-life	verdict
A	0.85	0.163	4.3 days	ACCEPT
B	0.95	0.051	13.5 days	ACCEPT
C	0.99	0.010	69 days	REJECT (too slow)
D	1.005	-	none ($\text{phi} \geq 1$)	REJECT (no revert)

Pairs A and B revert fast enough to trade. Pair C reverts, but far too slowly. Pair D has a coefficient at or above one - it doesn't revert at all.

Applying the Filter

Accept two, reject two

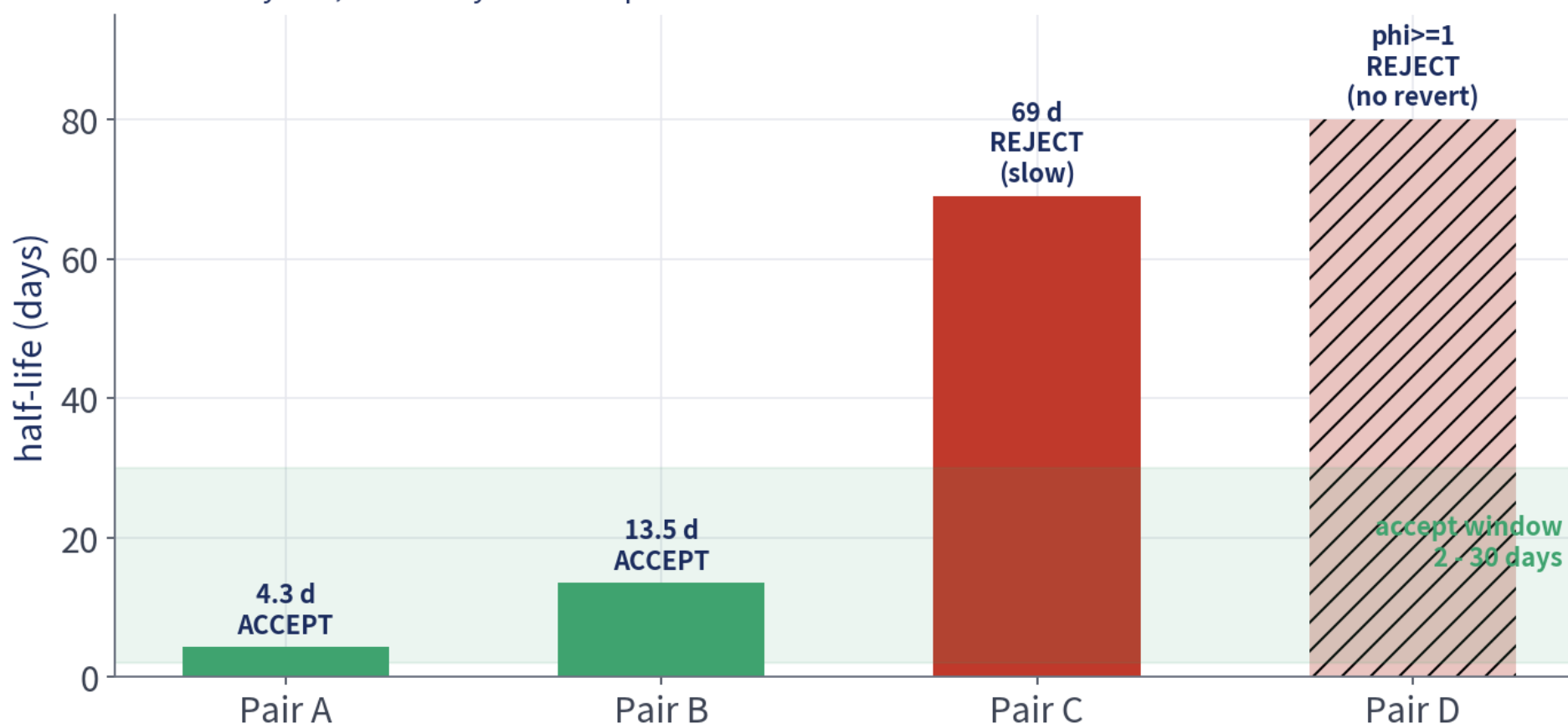
A ($\phi = 0.85$) : HL = 4.3 days $\Rightarrow$ **ACCEPT**
B ($\phi = 0.95$) : HL = 13.5 days $\Rightarrow$ **ACCEPT**
C ($\phi = 0.99$) : HL = 69 days $\Rightarrow$ **REJECT** (too slow)
D ($\phi = 1.005$) : $\phi \geq 1$ $\Rightarrow$ **REJECT** (not reverting)

The filter cleanly separates the portfolio: the two fast reverters go in, the capital trap and the non-stationary pair stay out.

Only Fast Reverters Pass

Pairs A and B fall inside the 2-to-30-day accept window and go into the portfolio. Pair C's 69-day half-life is a capital trap, and Pair D has a coefficient at or above one - no reversion at all. The half-life screens them in one glance

Case A: only fast, stationary reverters pass the half-life filter



Build the Portfolio Deliberately

The half-life is your first gate

Why screen at all

- Most candidate spreads aren't tradeable.
- Slow reverters lock up capital for thin returns.
- Non-stationary ones will just drift away.

What the filter buys you

- A portfolio of pairs that actually turn over.
- A single, comparable number per pair.
- A first line of defense against bad trades.

The Trade Economics

Why the half-life is the money number

Two pairs can offer the same profit per trade yet very different returns - because the faster one recycles capital more often. A round-trip takes a few half-lives, so a short half-life means more trades per year. We compute the annualized return for a fast and a slow reverter with the same 2% edge

The inputs

- Same 2% gross profit per round-trip.
- A round-trip runs about three half-lives.
- Pair A at 4.3 days, Pair B at 13.5.

What we'll do by hand

- Turn the half-life into trades per year.
- Multiply by the per-trade profit.
- Compare the two annualized returns.

Trades Per Year

Turnover comes from the half-life

$$\text{trades/yr} = \frac{250}{\text{cycle}} \approx \frac{250}{3 \times \text{half-life}} \quad (\text{faster reversion} \Rightarrow \text{more turns})$$

The cycle

- A trade reverts in about three half-lives.
- That's the capital-commitment time.
- Divide the trading year by it.

The consequence

- A short half-life means many trades a year.
- A long one means only a handful.
- Turnover is set by reversion speed.

Fast vs Slow, Annualized

The same edge, very different returns

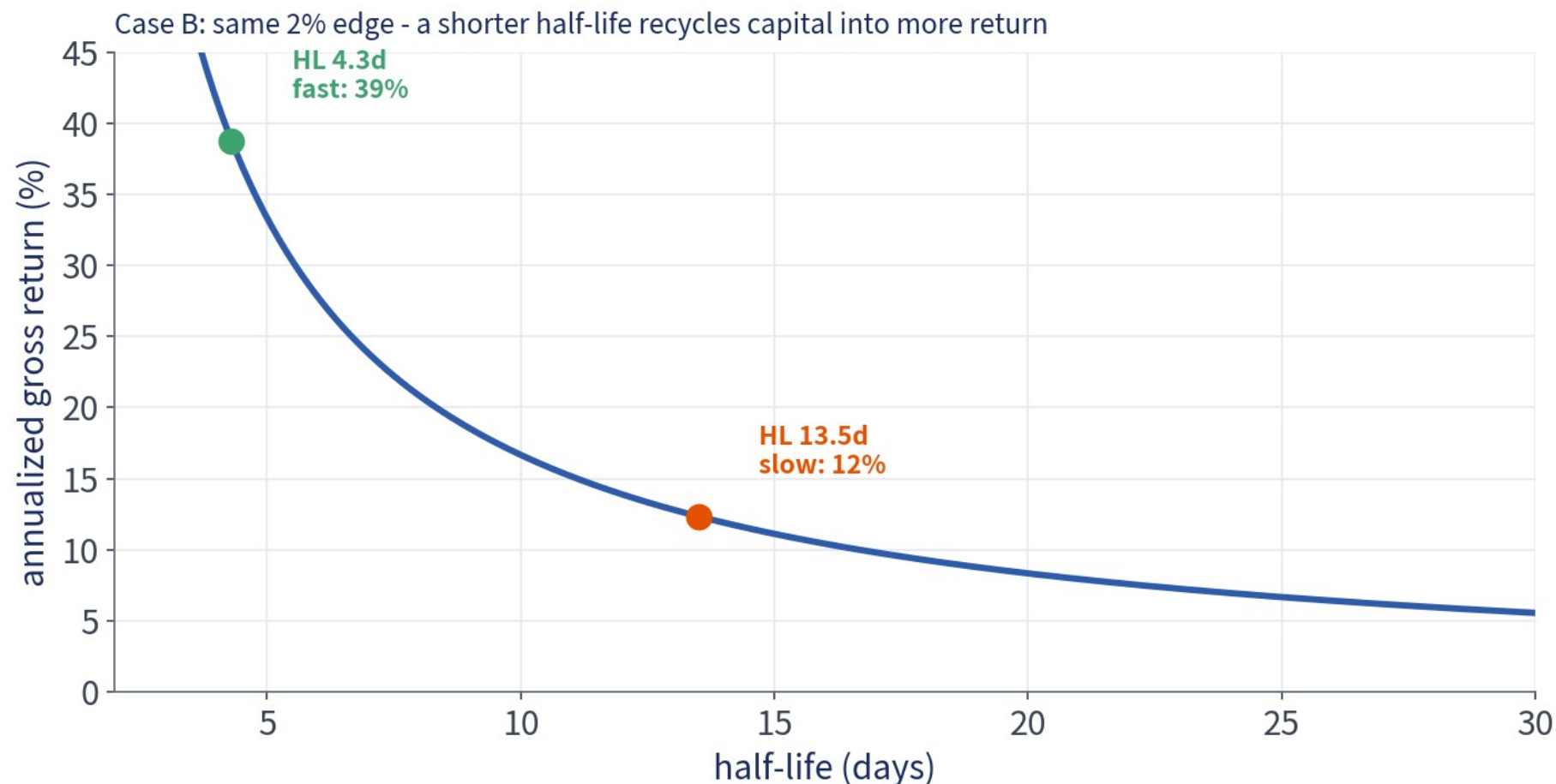
$$\begin{aligned} \text{fast (HL 4.3): } & \frac{250}{12.9} = 19.4 \text{ trades} \times 2\% = \mathbf{39\%} \\ \text{slow (HL 13.5): } & \frac{250}{40.5} = 6.2 \text{ trades} \times 2\% = \mathbf{12\%} \end{aligned}$$

Both capture 2% per trade, but the fast reverter turns capital 19 times a year for 39%, versus 6 times for 12% - roughly three times the return from the same edge.

Shorter Half-Life, More Return

With a fixed 2% edge, the annualized return falls off as the half-life lengthens - because capital turns over less often. The fast reverter at 4.3 days earns about 39%; the slow one at 13.5 days only 12%.

Reversion speed, not the per-trade edge, dominates



Turnover Beats Edge

And the risk that undoes it all

The economics

- Return is edge times turnover.
- The half-life sets the turnover.
- A faster reverter compounds more often.

The risk

- A regime break stops the reversion entirely.
- The spread drifts and the trade bleeds.
- Re-estimate constantly; a broken pair has no half-life.

The Pairs Journey, Now Five Stages

Volatility bands sized the trade (W5), cointegration anchored the spread (W7), a Kalman filter kept the hedge live (W9), a regime gate decided when to trade (W10), and now an OU model gives the spread a principled mean-reversion structure (W11). Stage V is principle over heuristics.

The Pairs Journey, Now Five Stages

Stage V replaces threshold heuristics with a principled OU model of the spread.



Stage V: principle over heuristics

The half-life screens pairs, the stationary band sets thresholds, and OU ties bands, holding time, and risk into one model.

One Number Runs the Portfolio

The half-life, everywhere

Stage V hangs on the half-life. It screens which pairs are worth trading, it sets how long each trade takes, and through turnover it drives the annual return. The OU model turns a spread's own statistics into a full trade specification - and reminds you to keep re-estimating because a broken spread has no half-life at all

Screening (Case A)

- Half-life filters the candidate pairs.
- Fast and stationary get in.
- Slow and non-reverting stay out.

Economics (Case B)

- Half-life sets trades per year.
- Turnover times edge is the return.
- Faster reversion, more compounding.

Reproduce the Screen and the Economics

Direct Claude Code to check your arithmetic

GOAL Verify the half-life screen and the trade economics, then build a small pair book.

[INPUT] Prompt · Claude Code

Write and run `part3_practice.py` that:

CASE A - screen:

- * for $\phi = 0.85, 0.95, 0.99, 1.005$, print the half-life and ACCEPT/REJECT (window 2-30 days, $\phi < 1$);

CASE B - economics:

- * with a 2% edge and cycle = 3 half-lives, print trades/year and annualized return for HL 4.3 and 13.5 days.

Then fit OU to several real spreads and rank them by half-life into a small pair book.

Match every printed number to the slides.

Check against the slides

- HL: 4.3, 13.5, 69 days.
- A, B accept; C, D reject.
- fast: 19 trades -> 39%.
- slow: 6 trades -> 12%.

Deliverable: `part3_practice.py` whose printed numbers match the screen and the economics.

Three Questions on the Cases

If you can answer these, the practice has landed

Q1 Why screen pairs by half-life?

**To keep only fast,
stationary reverters**

Q2 Why does a shorter half-life earn more?

**It recycles capital
into more trades a
year**

Q3 What kills an OU trade?

**A regime break - the
spread stops
reverting**

All three come from the half-life filter and the turnover economics.

Three Pitfalls to Avoid

Where Stage V slips

Common mistakes

- "Trade any spread that reverts eventually."
- "A bigger per-trade edge is all that matters."
- "Fit the half-life once and trust it forever."

The reality

- A slow reverter is a capital trap - screen it out.
- Turnover matters as much as edge for the return.
- Spreads break; re-estimate and stay humble.

Two Calculations to Remember

Part 3 applications in one screen

1

The half-life screens the portfolio

Of four candidates, A (4.3 days) and B (13.5 days) passed the 2-to-30-day filter; C reverted too slowly at 69 days, and D had a coefficient at or above one, so it doesn't revert at all. One number, four verdicts.

2

Turnover turns edge into return

With the same 2% per-trade edge, the fast reverter turned capital 19 times a year for 39%, while the slow one managed 6 turns for 12% - roughly three times the return, purely from a shorter half-life.

3

Stage V completes the pairs thread

Volatility bands, a cointegration anchor, a Kalman hedge, a regime gate, and now an OU model. The half-life screens pairs, sizes trades, and drives returns - principle in place of heuristics.

Next — Week 12: from a single stochastic process to systems, as the course turns to multivariate and machine-learning methods.